Warrant: Reproducible Evaluation and Evidence Management for AI Detectors

Aditya Singh
IIT Kharagpur

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Abstract

A detector score is not itself evidence that an operating decision satisfies a declared requirement. Warrant is a local Python toolkit connecting scored, labelled inputs to explicit decisions, retained records and arithmetic replay. This report describes its architecture and investigates the scope of its empirical and statistical claims. Retained benchmark results show within-envelope error-label discrimination alongside low recall at a restrictive threshold, and a constructed context-transfer experiment in which a proposed pooling strategy did not remain robust. Other artifacts cover configuration, declared-workload sampling, monitoring simulations and bounded audit studies. A source-connected investigation finds that all 92 quantitative README rows match retained fields across 21 artifacts, while 15 rows point directly to dirty-source records. A separately implemented rank/count oracle agrees on 25 frozen-score targets, directly covering 19 README point claims. A preserved tied-score counterexample also limits the general interpretation of the native adaptive returned false-positive bound. These observations support an inspectable evaluation and evidence-management artifact, not universal assurance, independently adjudicated labels or production validation.

Keywords: detector evaluation; reproducibility; evidence provenance; statistical interpretation; operating points; monitoring.

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1 Introduction

A detector can rank suspicious items without supporting an application decision. Its score does not specify the clean items that will be flagged, positive cases that will be missed, or conditions under which an observed rate transports to another population. An evaluation therefore needs an estimand, labelled evidence, an operating rule, uncertainty and a record of selection.

Warrant implements a local workflow around those requirements. It reads an existing scored file, evaluates a declared decision, retains structured evidence and allows material calculations to be replayed. The repository also contains probe research, distribution-shift experiments, challenge-set measurements and studies of sampling, monitoring and evaluation accounting.

This report asks:

1. What does the implemented software retain and verify?
2. What do its experiments demonstrate, including adverse outcomes?
3. Which stronger interpretations are not supported by retained evidence?

The contribution is an inspectable systems artifact and an evidence-centered technical account: a working local pipeline, a source-connected discussion of bounded measurements, and an internal audit distinguishing field agreement, separate point arithmetic, source lineage and statistical

validity. It does not claim a new neural architecture, binomial theorem, complete monitoring theory or comprehensive novelty review.

The audit is deliberately self-critical. Favorable results cannot erase an unrecorded producing state, inappropriate independent unit or stronger guarantee than an implemented method supports. Preserving those limitations is part of the result.

2 Related work and positioning

Internal-state classifiers have been studied as predictors of truth-related labels [1]. Warrant's probe experiment is narrower: it predicts a recorded matching-rule outcome from a prompt state before generation. It does not identify belief, deception or a universal truth signal.

Exact-binomial intervals [3], Neyman–Pearson classification [10], and bootstrap methods [5] are established foundations. Their use here is engineering application, not theoretical novelty. The sampling unit, selection procedure and claimed guarantee must still match.

Marginal, training-conditional and parameter-confidence statements are distinct [2, 4]. The returned-bound diagnostic below concerns a specific interpretation in this implementation, not a blanket invalidity claim about conformal prediction.

Time-uniform inference and e-detectors also have an established literature [6, 9]. The repository applies and simulates particular rules; favorable finite-horizon findings do not validate every implementation.

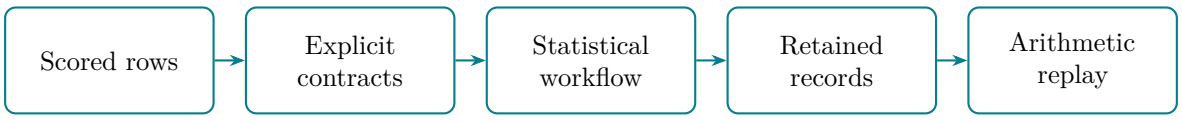
The grader and agent-accounting tracks retain separate closest-work assessments and stopping decisions. Their inclusion is not evidence of a publication-ready method contribution. This manuscript is a technical preprint and makes no venue-acceptance claim.

3 System design

3.1 Contracts, calculations and publication

The customer workflow consumes `id,label,score` records with a declared positive condition and score direction. Optional groups and second labels carry additional structure. The reader validates syntax and supported values; it cannot establish a scientifically correct rubric from arbitrary columns.

The architecture separates input contracts, calculation, publication and replay (Figure 1). A malformed file is an input error. Insufficient evidence can produce a normal refusal. A changed record can fail replay. Those outcomes should not collapse into one generic process-success indicator.



External assumptions: label truth, sampling design, deployment identity and authentic origin.

Figure 1: Logical layers of the local toolkit, not a deployed multi-tenant service. Publication correctness and replay do not authenticate external assumptions.

3.2 Different records answer different questions

Preparation retains an explicit source mapping. Native certification selects an operating point for a requested budget. A frontier retains a declared grid and selection evidence. Ordinary rechecks use fresh batches; chains additionally retain a monitoring horizon and sequence prefix. Release registration instead freezes an already chosen threshold, criteria, identity and holdout membership.

Local publication distinguishes complete artifacts from interrupted writes and coordinates chain continuation. Hashes detect changes relative to retained reference material. They do not supply an external signing authority, trusted time or proof that no alternate private history exists.

GPU extraction is separate from the ordinary customer path. Fitting and study code can later use caches or frozen scores. The small customer environment does not need to load a language model to evaluate an existing scored file.

4 Data and methods

4.1 Recorded question-answer data

The question source is TriviaQA [7]. The retained main collection contains 2400 selected questions, partitioned into 1200 training, 600 validation and 600 test rows. The recorded model is Qwen2.5-7B-Instruct [8], under the manifest's generation and quantization settings.

The positive label marks a response that fails the project's matching rule. Lenient alias matching is an operational reference rather than independent human adjudication. Strict matching supplies sensitivity information, not an automatically correct replacement.

The last-token feature is the final real prompt position *before* generation. The classifier standardizes features and fits logistic regression on training indices; regularization selection uses the declared validation split. Later project analyses reuse the collection, so not every subsequent use is described as pristine confirmation.

The model/tokenizer loader does not pass an explicit immutable revision. Its provenance method records model name and configuration facts rather than a complete historical weight hash. Model-level reconstruction needs more than a later download of the same identifier.

4.2 Constructed transfer and challenge data

The context-transfer experiment changes prompt context for feature extraction while reusing the original questions, short-context answers and labels. It measures representation transfer relative to fixed outcomes, not newly generated long-context answer accuracy.

The PII studies use authored scenarios, systematic forms and synthetic identifiers. A companion variant set extends forms beyond the base inventory. Positive enrichment and shared construction limit transport and independence.

The benign challenge repeats scenarios in different framings. Historical row-binomial intervals remain replayable but are not current independent-unit assurance. Selected canaries are regression tripwires, not independent quality evidence.

4.3 Quantities and uncertainty

At a fixed threshold,

$$\text{recall} = \frac{TP}{TP + FN}, \qquad \text{precision} = \frac{TP}{TP + FP}, \tag{1}$$

$$\text{FPR} = \frac{FP}{FP + TN}, \qquad \text{flag rate} = \frac{TP + FP}{N}. \tag{2}$$

AUROC is a positive-versus-negative pair-ranking probability with half credit for ties. Lift is recall divided by realized flag rate on the same population, equivalently precision divided by prevalence.

These quantities have different denominators. Precision is undefined when nothing is flagged; ranking metrics are withheld on a single-class challenge. Research intervals retain their bootstrap count, seed, grouping and method. Arithmetic reconstruction does not establish their coverage.

For a fixed rule and independent negative trials, the one-sided exact-binomial upper bound is

$$U(k, n, \delta) = \begin{cases} \text{Beta}^{-1}(1 - \delta; k + 1, n - k), & k < n, \\ 1, & k = n. \end{cases} \tag{3}$$

Selection and dependence require additional arguments.

4.4 Requested budget and returned number

Under an appropriate independently fixed scorer and IID calibration design, an order-statistic procedure can target a fixed-budget proposition

$$\Pr_D\{R(t(D)) > B\} \leq \delta. \tag{4}$$

Complete independent review of the implementation/assumption mapping remains outstanding. The proposition is different from a general claim about a data-dependent numerical return:

$$\Pr_D\{R(t(D)) > U(D)\} \leq \delta. \tag{5}$$

Section 6.2 gives a preserved witness against Equation 5, without a violation of Equation 4 in that specified distribution.

Registered release evaluation has a fixed threshold before the holdout. Its implementation allocates error across five rate intervals and applies the registered FPR/recall acceptance criteria. The union bound does not require independence between interval events; each binomial calculation still needs the appropriate IID design. Grouped release input is unsupported.

4.5 Independent point audit

The audit parses every current README-table row, resolves its field uniquely, checks stated precision and records artifact hashes and source metadata. A reviewed family registry connects data references, writers, interpretations and reproduction boundaries.

A separate oracle computes confusion counts and tied-pair AUROC. For each positive score, binary searches in sorted negative scores count strict wins and ties. It does not use the original metric-builder formulas to obtain those points. This is calculation independence, not an independent investigator or label source.

Missing classifications, invalid score evidence and source changes during receipt generation are refused. Original scientific artifacts are never rewritten.

5 Retained findings

5.1 Discrimination and operating trade-offs

The main held-out AUROC is 0.8256, with recorded interval [0.7934, 0.8567]. At its operating point, recall is 0.0794, precision 0.8800 and flag rate 0.0417 (Table 1).

The independently checked block has 277 positive rows among 600 total rows, with 25 flagged and 22 true positives. High ranking association therefore coexists with many missed positive rows at that cut. Different targets explain the coexistence arithmetically; AUROC does not guarantee application usefulness.

Table 1: The main operating point. Intervals come from retained artifacts; the point oracle does not independently validate their coverage. Claim IDs identify the full source ledger.

Quantity	Value	Recorded interval	Claim
AUROC	0.8256	[0.7934, 0.8567]	C001
Recall	0.0794	[0.0496, 0.1115]	C002
Precision	0.8800	[0.7500, 1.0000]	C003
Flag rate	0.0417	[0.0267, 0.0583]	C004

The broader budget sweep illustrates a trade-off, but its largest-favorable-budget headline is selected from an observed grid. The validation-selected/test-scored sweep adds split separation while retaining dirty-source provenance. Neither creates a fresh external confirmation collection.

5.2 The pooling rescue did not hold

In the constructed context transfer, last-token AUROC remains 0.8135, mean pooling is near chance at 0.5015, and rolling maximum is 0.5553 (Table 2). The latter flags 0.5433 of target rows.

Table 2: Constructed feature transfer. Questions, original responses and labels are reused; this is not fresh long-context generation accuracy.

Pooling	Target AUROC	Recorded interval	Claim
Last token	0.8135	[0.7797, 0.8447]	C029
Mean	0.5015	[0.4546, 0.5479]	C030
Max rolling means	0.5553	[0.5105, 0.6015]	C031

Global averaging may dilute a local feature; maxima over many windows may emphasize distractor extremes. Position and general feature-distribution changes are other possible mechanisms. The experiment does not isolate them. The rolling-window motivation remains a hypothesis that did not establish the predicted robust alternative.

A stronger mechanism study would predeclare interventions on distractor length, position and content, keep the source fit fixed, and confirm on independent questions. That study was not performed.

5.3 Configuration, planning and monitoring

Stock PII recall is 0.1176 and the custom-augmented configuration's recall is 0.6176 on the authored challenge. This supports configuration coverage as important within that inventory.

Custom patterns were informed by base forms, the companion set shares construction choices, and enrichment makes precision population-specific. Broad superiority to current upstream systems is not established.

For confirmed positives TP, unflagged volume N_u and unflagged error prevalence q, recall on traffic satisfies

$$R(q) = \frac{TP}{TP + qN_u}, \qquad \frac{\partial R}{\partial q} = -\frac{TPN_u}{(TP + qN_u)^2}. \tag{6}$$

The declared workload's derivative is -4.6496. Sizing prevalence directly in recall units targeted the wrong quantity. The retained first-month plan is 1399 labels under its assumptions; the superseded 187 remains failure evidence. Discrete, finite-population and assurance calculations also matter; the derivative alone is not the headline rule.

Coverage 0.9680 concerns 1000 simulated trials. No customer labeling process or commercial saving was measured. Similarly, the year-long repeated-fixed-test and chained S0 cells report 0.8515 and 0.0085 simulated revocation shares, conditional on their rules and horizon. The historical capped-monitor cell is not a measurement of the later repaired runtime.

5.4 Finite demonstrations and closed extensions

The gate retains 33 selected detector-independent comparisons and blocks 2 of 8 authored legitimate workflows. These are finite-set counts, not production robustness or false-block rates.

The score-free experiment uses conditionally independent simulated votes. Its results are an optimistic model-specific scenario, not a universal upper bound for real judges.

The later frozen grader-discovery extension found no new omission under its bounded method, despite accepted controls. The unscored-trial work closed as a technical report: strict-cohort agreement is partly an eligibility condition, and no original execution roster or corrected official ranking is recovered. Stopping-feedback closed at data feasibility; no policy effect was estimated. These outcomes remain part of the evidence.

6 Source audit and assurance limits

6.1 Matching values does not reconstruct source

All 92 numerical rows match their retained fields across 21 artifacts (Table 3). This is not independent regeneration of every result.

Table 3: Final source-audit scope, generated from the retained receipt.

Check	Count	Meaning
Matching README rows	92	Agreement at stated precision
Result artifacts	21	Files directly named by the table
Reviewed families	18	Shared method and source context
Dirty-source rows	15	Producing state not fully identified
Frozen-score point targets	25	Separate rank/count arithmetic agrees
Directly backed README points	19	Subset linked to those targets

Dirty-source rows C006–C007, C016–C019 and C020–C028 concern reconciliation, the out-of-sample sweep and resplit policy results. A dirty flag may reflect an unrelated untracked file or a substantive edit. Its metadata does not establish which, so the named commit alone cannot reconstruct the complete original state.

Downstream use requires the same context even when a later writer was clean. Relabeling an old result under a clean later commit would hide the gap. An archived original state or a separately identified new study is required.

The point oracle directly backs 19 current table claims. Other rows include counts, statuses, calculations, simulations and measurements with different reproduction requirements. Standard frozen-score replay also reconstructs supported intervals using their historical method. That is not independent interval validation or fresh warrant issuance.

6.2 Controlled returned-bound counterexample

The preserved diagnostic sets $n = 100$, $B = 0.05$, $\delta = 0.05$ and design count $K = 1$. IID scores in $\{0, 1, 2\}$ have tail probabilities $q_1 = 0.04656$ and $q_2 = 0.029514$.

Let M_1, M_2 count scores at least 1 and 2. Under the preserved successor-threshold convention, returned-bound noncoverage is the disjoint union

$$\{M_1 \leq 1\} \ \cup \ \{M_1 \geq 2, \ M_2 = 0\}. \tag{7}$$

Its probability is

$$(1 - q_2)^n + nq_2(1 - q_1)^{n-1} \approx 0.076302115 > 0.05. \tag{8}$$

The exact program enumerates 5151 multinomial count states and checks the closed form with rational arithmetic.

Every selected population risk in this example is $0, q_1$ or q_2, below B. The witness challenges the general adaptive returned-U interpretation without a requested-budget violation in that distribution. It is a known internal witness reproduced arithmetically, not a new general theorem or a completed independent production review. FPR-001 remains open.

6.3 Scope and terminology

The matrix includes synthetic fixtures and historical method records. A VALID-cell count is not independently validated model coverage. Historical dependent-row intervals remain historical despite successful replay.

The vote simulation is likewise an optimistic independent-vote scenario rather than a universal upper bound. Narrower wording can be scientifically necessary even when no measured value changes.

7 Discussion

The favorable and adverse results have a common implication: evaluation must retain the proposition that its evidence can actually support. Ranking, operating recall, future FPR, simulated catch probability and exact workflow counts cannot share an interpretation merely because they appear together.

A separate calculation can strengthen arithmetic confidence without becoming an independent experiment. A checksum strengthens byte identity without authenticating an author. A clean source record improves reproducibility without proving labels or sampling. An explicit refusal can be a correct outcome.

The supported contribution is a locally executable evaluation instrument and a research record whose assumptions can be inspected. This report adds an organized source-connected account of results and limits. Presentation quality and code volume are not substitutes for novelty, external validation or production utility.

8 Threats to validity

Labels and sampling. Alias matching, authored challenges, enrichment and repeated forms limit interpretation and transport. Independent blinded adjudication was not completed. Model pretraining exposure to individual questions was not audited.

Selection and reuse. Project phases reuse the corpus. Some sweeps are exploratory and some records have dirty-source provenance. A later declared partition cannot erase earlier exposure.

Reconstruction. Original extraction caches and complete immutable model identity are not all committed. Score replay is stronger than reading a table but weaker than reproducing model behavior.

Inference. Bootstrap coverage, independent units, conditional nulls and adaptive selection need their assumptions. FPR-001 and legacy grouped-method limits remain open. Registered release evaluation has a different design and must not inherit an unrelated guarantee.

Internal review. The oracle is separately implemented arithmetic within the same project, not an external investigator. Automated assistance was used in implementation and documentation. Independent human scientific sign-off is not claimed.

Operation. No production deployment, authenticated evidence service, buyer study or commercial benefit is established. Finite demonstrations and simulated years do not replace those observations.

9 Reproduction and availability

The prepared distribution contains the PDF, standalone LaTeX source and a tracked source/evidence snapshot. Code and third-party notices remain in that snapshot. Untracked private review material and ignored extraction caches are excluded.

The development repository was private during preparation. Public reproduction therefore requires distribution of the accompanying snapshot or access to the necessary source and inputs. A PDF-only deposit is not a complete reproduction package. No DOI or publication acceptance is invented by the build.

The main CPU checks are:

```
python scripts/verify.py --no-activations
python scripts/audit_documentation.py --links
python -m pytest tests/ -q
```

New receipts require clean committed source and a new output destination. Original extraction replay has additional input requirements and is reported separately; a skipped activation tier is not a pass.

The LaTeX sources use a declared Tectonic toolchain and BibTeX bibliography. The paper guide records compilation and deposit preparation. Numeric macros are generated from retained evidence. Typesetting is not a new model experiment.

10 Conclusion

Warrant connects local operating decisions to retained evidence and explicit replay. Its experiments demonstrate useful correlations and important failure modes. The source audit confirms numerical agreement and separate point arithmetic, while exposing source gaps and an unresolved guarantee interpretation. The defensible result is a reproducible, bounded technical artifact with a transparent account of its limits. Stronger scientific, operational and publication claims require additional evidence.

A Evidence map

The complete per-claim ledger and qualitative register accompany the source. These families are not one homogeneous experiment.

Family	Question	Evidence and boundary
F01–F03	Probe and paired comparisons	Validation, reconciliation and paired artifacts; recorded labels and historical source conditions apply.
F04–F06	Budgets and profiles	Exploration and validation/test calculations; dirty provenance affects the out-of-sample and resplit-policy artifacts.
F07–F09	Transfer, drift and matrix	Constructed feature shift and replay; the inventory includes fixtures.
F10–F11	PII and benign challenges	Authored cases and synthetic identifiers; legacy dependent-row inference is not current assurance.
F12–F13	Feasibility and label planning	Declared-scenario arithmetic and simulation, without production cost measurement.
F14–F15	Monitoring	Scenario- and horizon-specific simulations; historical runtime distinction retained.
F16	Label sensitivity	Hypothetical label perturbations; disagreement is not independent truth.
F17	Score-free voting	Conditionally independent simulated votes, not commercial-judge measurements.
F18	Action gate	Finite declared workflows and states, not general adversarial robustness.

B Audit and manuscript provenance

The original consolidated system baseline is `9dbdee03e77b698c497186b9b00297ee506cd0fe`. The final numerical audit records producing source:

```
c789c9514a1cdba09de63d0c77cd51a593837ba3
```

Its JSON retains selectors, raw hashes, source flags and explicit exclusions.

Initial and final guarded receipts are both retained. Guard refinements check source stability and resolve equivalent selectors by object identity rather than numerical equality. They do not rewrite scientific measurements.

The manuscript source includes metadata, numeric generation, bibliography and compiler pins. The archival manifest identifies the accompanying source snapshot. Publishing a Zenodo preprint is a separate author action and does not imply peer review.

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